2 EXAM PRACTICE

1 The relative atomic mass of boron is 10.8.

A sample of boron contains the isotopes $^{10}_{5}B$ and $^{11}_{5}B$.
What is the percentage of $^{11}_{5}B$ atoms in the isotopic mixture of this sample?

A 0.8% **B** 8.0% **C** 20% **D** 80% [1]

(Total for Question 1 = 1 mark)

2 Which of the following elements has no paired p electrons in a single uncombined atom of the element?

A carbon **B** oxygen **C** fluorine **D** neon [1]

(Total for Question 2 = 1 mark)

3. Which of the following electronic configurations is that of an atom of an element which forms a simple ion with a charge of −3?

A $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$ **B** $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

C $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1\ 4s^2$ **D** $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2$ [1]

(Total for Question 3 = 1 mark)

4 A sample of chlorine contains isotopes of mass numbers 35 and 37.

The sample is analysed in a mass spectrometer. How many peaks corresponding to Cl_2^+ are recorded?

A 1 **B** 2 **C** 3 **D** 4 [1]

(Total for Question 4 = 1 mark)

5 What is the atomic number of an element that contains atoms which have four unpaired electrons in their ground state?

A 6 **B** 16 **C** 22 **D** 26 [1]

(Total for Question 5 = 1 mark)

6 Which of the following ions has more electrons than protons, and also has more protons than neutrons?

($H = ^{1}_{1}H$ $D = ^{2}_{1}H$ $He = ^{4}_{2}He$ $O = ^{16}_{8}O$)

A OD^- **B** D_3O^+ **C** He^+ **D** OH^- [1]

(Total for Question 6 = 1 mark)

7 A sample of helium from a rock was found to contain two isotopes with the following composition by mass: ^3He, 0.992%; ^4He, 99.008%.

(a) State what is meant by isotopes. [1]

(b) State the difference in the atomic structures of ^3He and ^4He. [1]

(c) (i) Which isotope is used as the basis for relative atomic mass measurements? [1]

(ii) Calculate the relative atomic mass of helium in the rock sample. [2]

(d) Helium has the largest first ionisation energy of all the elements.

(i) State what is meant by first ionisation energy. [2]

(ii) Write an equation, including state symbols, to represent the first ionisation energy of helium. [2]

(iii) Explain why the first ionisation energy of helium is larger than that of hydrogen. [2]

(Total for Question 7= 12 marks)

8 The five ionisation energies of boron are:

801 2427 3660 25 026 32 828

(a) State and justify the group in the Periodic Table in which boron is placed. [2]

(b) Which of the following represents the **second** ionisation energy of boron?

A $B(g) \rightarrow B^{2+}(g) + 2e^- \quad \Delta H = +2427\ kJ\ mol^{-1}$

B $B^+(g) \rightarrow B^{2+}(g) + e^- \quad \Delta H = +2427\ kJ\ mol^{-1}$

C $B(g) \rightarrow B^{2+}(g) + 2e^- \quad \Delta H = -2427\ kJ\ mol^{-1}$

D $B^+(g) \rightarrow B^{2+}(g) + e^- \quad \Delta H = -2427\ kJ\ mol^{-1}$ [1]

(c) Explain why the second ionisation energy of boron is larger than the first. [2]

(d) Give the electronic configuration of a boron atom. [1]

(e) Is boron classified as an s-, a p- or a d-block element? Justify your answer. [2]

(f) Explain why the first ionisation energy of boron is less than that of beryllium, even though a boron atom has a greater nuclear charge. [2]

(Total for Question 8 = 10 marks)

9 (a) State what is meant by periodic property. [1]

(b) Explain the **general** trend in first ionisation energy of the elements Na to Ar in Period 3 of the Periodic Table. [3]

(c) Explain the trend in atomic radii of the elements sodium to chlorine in Period 3 of the Periodic Table. [4]

(d) The table shows the melting temperatures of the elements sodium to chlorine in Period 3 of the Periodic Table.

Symbol of element	**Na**	**Mg**	**Al**	**Si**	**P**	**S**	**Cl**
melting temperature / °C	98	649	660	1410	44	113	−101
bonding							
structure							

Complete the table using the following guidelines.

(i) Complete the 'bonding' row using **only** the words 'metallic' or 'covalent'.

(ii) Complete the 'structure' row using **only** the words 'simple molecular' or 'giant lattice'.

(iii) Explain why the melting temperature of phosphorus is different from that of silicon. [5]

(Total for Question 9 = 13 marks)

10 (a) Complete the table to show the properties of the three major sub-atomic particles. [3]

Particle	Relative charge	Relative mass
proton		
neutron		
electron		

(b) The particles in each pair below differ **only** in the number of protons **or** the number of neutrons **or** the number of electrons that they contain. State the difference in each pair.

(i) ^{16}O and ^{17}O

(ii) ^{24}Mg and $^{24}Mg^{2+}$

(iii) $^{9}K^{+}$ and $^{40}Ca^{2+}$ [3]

(Total for Question 10 = 6 marks)

11 (a) State what is meant by an orbital. [2]

(b) Draw the shapes of:

(i) an s-orbital

(ii) a p-orbital. [2]

(c) How many electrons can occupy

(i) an s-orbital

(ii) a p-subshell? [2]

(d) Using s and p notation, give the electronic configurations of each of the following atoms and ions:

(i) Na

(ii) O^{2-}

(iii) Mg^{2+}

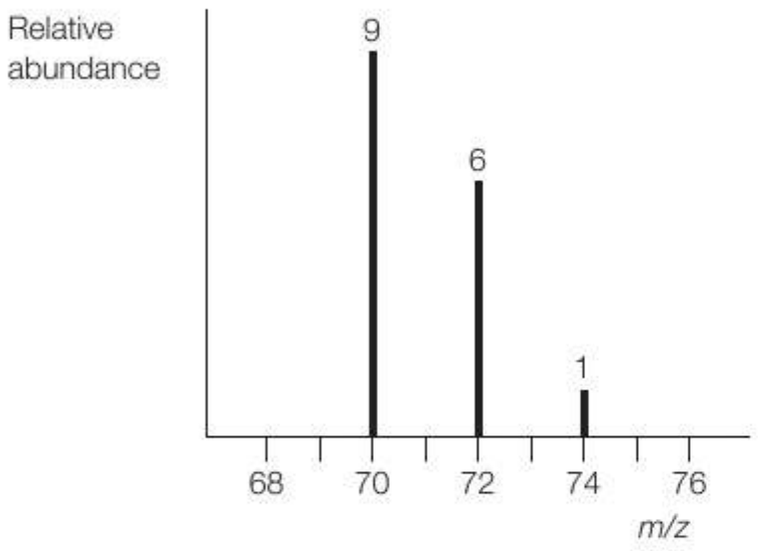
(iv) Cl [4]

(e) Which of these ions contains the fewest number of electrons?

A NH_4^+ **B** P^{3-} **C** S^{2-} **D** Cl^- [1]

(Total for Question 11 = 11 marks)

12 Chlorine has two isotopes, ^{35}Cl and ^{37}Cl.
The diagram shows part of the mass spectrum of a sample of chlorine gas.



(a) Give the formula of the species responsible for each peak at m/z of 70, 72 and 74. [3]

(b) Give the m/z ratio for the two other peaks in the mass spectrum of chlorine. Give the formula of the species responsible for each peak and state the relative abundance of these two species. [5]

(Total for Question 12 = 8 marks)

13 The table gives the first four ionisation energies of the elements sodium, magnesium and aluminium.

Ionisation energy/kJ mol^{-1}				
Element	**1st**	**2nd**	**3rd**	**4th**
sodium	496	4563	6913	9544
magnesium	738	1451	7733	10 541
aluminium	578	1817	2745	11 578

Explain why:

(a) the first ionisation energy of sodium is lower than that of the first ionisation energy of magnesium. [2]

(b) the first ionisation energy of magnesium is higher than the first ionisation energy of aluminium. [2]

(c) the second ionisation energy of magnesium is lower than the second ionisation energy of aluminium. [2]

(d) the fourth ionisation energy of aluminium is higher than its third ionisation energy. [2]

(Total for Question 13 = 8 marks)

14 Lithium is an element in the s-block of the Periodic Table. Naturally occurring lithium contains a mixture of two isotopes, ^{6}Li and ^{7}Li.

(a) Complete the table to show the atomic structure of each of the two isotopes.

Isotope	Number of portions	Number of neutrons	Number of electrons
^{6}Li			
^{7}Li			

[2]

(b) A sample of lithium was found to contain 7.59% of ^{6}Li and 92.41% of ^{7}Li.

(i) Give the name of the instrument that can be used to determine this information. [1]

(ii) The ^{7}Li isotope has a relative isotopic mass of 7.016 and the ^{6}Li isotope has a relative isotopic mass of 6.015. State what is meant by relative isotopic mass. [2]

(iii) Use the relative isotopic masses to calculate the relative atomic mass of lithium. Give your answer to three significant figures. [2]

(c) (i) Give the electronic configuration of an atom of lithium. [1]

(ii) State why lithium is described as an s-block element. [1]

(iii) Would you expect the two isotopes of lithium to have the same chemical properties? Justify your answer. [1]

(Total for Question 14 = 10 marks)